

DEPARTMENT OF COMPUTER ENGINEERING
FACULTY OF ENGINEERING,
UNIVERSITY OF JAFFNA
EC9570 – DIGITAL IMAGE PROCESSING
ASSIGNMENT

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1. Introduction

As part of the Smart Supermarket Product Identification System, a deep learning-based object detection model was trained to automatically detect and classify grocery products from shelf images. The goal of this stage of the project was to build a model capable of recognizing a large number of distinct retail products with high accuracy and low inference latency, so that it can later be integrated into the full detection-and-reporting pipeline.

2. Dataset Preparation and Split

The model was trained using the RPC (Retail Product Checkout) dataset, a large-scale benchmark dataset designed for retail product recognition. For this training run, a 6,000-image subset (the val2019 portion of the dataset) was used as the source data, since it provided a manageable yet sufficiently large sample for the available compute resources.

To create the training and testing sets, the 6,000 images were randomly divided using a fixed random seed (seed = 42) to ensure the split is reproducible. This resulted in:

- 4,000 images allocated for training
- 2,000 images allocated for testing

A verification step confirmed there was zero overlap between the training and testing image sets, ensuring the model's reported performance reflects genuine generalization rather than memorization.

The dataset's original annotations were provided in COCO JSON format. Since the YOLO training framework requires a different annotation structure, all annotations were converted from COCO format into YOLO format (normalized bounding box coordinates per image, stored as per-image label files). After conversion:

- The training set contained 48,895 bounding box annotations across the 4,000 images
- The testing set contained 24,707 bounding box annotations across the 2,000 images

A data.yaml configuration file was generated to define the dataset paths and class list for the training framework.

3. Product Categories

The dataset covers 200 individual product classes, which can be grouped into the following broader product families for easier interpretation:

- Puffed food
- Dried fruit
- Dried food

- Instant drink
- Instant noodles
- Dessert
- Drink
- Alcohol
- Milk
- Canned food
- Chocolate
- Gum
- Candy
- Seasoner
- Personal hygiene
- Tissue
- Stationery

Each family contains multiple individual SKU-level classes (for example, "1_puffed_food" through "12_puffed_food"), meaning the model must distinguish not just between broad categories but between visually similar variants within the same category.

4. Model Architecture and Training Configuration

The object detection model used was YOLOv8n (the "nano" variant of YOLOv8), chosen for its lightweight architecture and fast inference speed, which makes it well suited for real-time or near-real-time supermarket checkout applications. The model consists of approximately 3.38 million parameters across 130 layers.

Training configuration:

- Epochs: 30
- Image size: 640×640
- Batch size: 16
- Optimizer: AdamW (automatically selected), with a learning rate of approximately 4.9×10^{-5}
- Hardware: $2 \times$ NVIDIA Tesla T4 GPUs

- Total training time: approximately 35 minutes (0.576 hours)

The model was initialized from pretrained YOLOv8n weights and fine-tuned on the RPC training set, with validation performed on the held-out test set after every epoch.

5. Training Progress

Model performance improved consistently across the 30 training epochs. At the start of training, validation mAP@50 was close to zero, reflecting the untrained state of the classification head (which had to be adapted from 80 generic COCO classes to the 200 RPC product classes). By the final epoch, mAP@50 had risen to 0.918, with mAP@50-95 reaching 0.779. The upward trend in accuracy was still ongoing at epoch 30, suggesting that additional training epochs could yield further improvement.

6. Final Results

The final trained model (best.pt) was evaluated on the 2,000-image test set, with the following results:

Metric	Value
Precision	0.863
Recall	0.872
mAP@50	0.918
mAP@50-95	0.779

These results indicate that the model is able to both accurately localize products (high precision) and reliably detect most products present in an image (high recall), while maintaining a fast enough inference speed for practical deployment.

7. Conclusion and Next Steps

The trained YOLOv8n model successfully learned to detect and classify 200 retail product categories with an overall mAP@50 of 0.918, demonstrating strong feasibility for the Smart Supermarket Product Identification System. Building on these results, the trained model was used to develop an end-to-end detection and classification pipeline, which has since been integrated into a Point of Sale (POS) system, enabling automatic product recognition at checkout without manual barcode scanning